YOUNGJUN LEE, PH.D.

CONTACT INFORMATION

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RESEARCH INTERESTS

(1) Large-Scale Wearable and Population Health Data Analytics on Physical Activity and Mental Health Outcomes, with an Emerging Focus on Autism Spectrum Disorder; (2) Immersive Virtual Reality Exercise Interventions Leveraging Virtual Coach-Induced Social Presence to Alleviate Depressive Symptoms in Autism Spectrum Disorder

EDUCATION

[2025] **Michigan State University (Ph.D.)** Michigan, United States
Department of Kinesiology, Physical Activity in Youth Disabilities Lab
Dissertation: *"Immersive Virtual Reality Exercise to Improve Depressive Symptoms in Young Adults with Autism Spectrum Disorder"*
Advisor: Dr. Janet Hauck

[2022] **Korea University (M.S.)** Seoul, Korea
Department of Physical Education, Exercise Nutrition Biochemistry Lab
Dissertation: *"Effect of Physical Activity and Nutritional Behavioral Change on Metabolic Syndrome in Overweight Young Women of Single-person Households"*
Advisor: Dr. Jonghoon Park

[2020] **Korea University (B.Ed.)** Seoul, Korea
Department of Physical Education

PROFESSIONAL EXPERIENCE

[2026 – Present] **Postdoctoral Research Fellow**, Children's Hospital of Philadelphia, Department of Pediatrics, Division of Gastroenterology, Hepatology and Nutrition

[2026 – Present] **Affiliated Postdoctoral Researcher**, University of Pennsylvania, Annenberg School for Communication, Annenberg Extended Reality Lab

REFEREED JOURNAL PUBLICATIONS

Published

[P14] **Lee Y**. Resting Heart Rate Variability and the Persistence of Simulator Sickness During Virtual Reality Exercise in Young Adults with Autism Spectrum Disorder: A Proof-of-Principle Study. *Research in Autism.* In press.

[P13] **Lee Y**, Ciaccioni S, Stodden D, Hauck J, Lima RR, Pesce C. Physical Activity and Adolescent Mental Health Across 45 Countries: The Moderating Role of Country-Level Perceived Health. *Mental Health and Physical Activity*. In press.

[P12] **Lee Y.** Differential Moderation by Aerobic and Muscle-Strengthening Physical Activity of the Violence Victimization-Suicidal Ideation Association in Korean Adolescents. *Suicide and Life-Threatening Behavior.* 2026; 56:e70134. *DOI:* https://doi.org/10.1111/sltb.70134

[P11] Ciaccioni S, **Lee Y**, Capranica L, Urban A, May R, Massini S, Guidotti F. Building resilience and equity: An umbrella review of evidence for crisis management in grassroots sport. *Societies*. 2026;16(4):118. DOI: https://doi.org/10.3390/soc16040118

[P10] **Lee Y**, Guidotti F, Capranica L, Pesce C, Benzing V, Hauck J, Ciaccioni S. Olympic combat sports and mental health in children and adolescents with disability: A systematic review of controlled trials. *Frontiers in Psychology. 2025;16:1567978.* DOI: https://doi.org/10.3389/fpsyg.2025.1567978

[P9] **Lee Y**, Capranica L, Pesce C, Guidotti F, Benzing V, Hauck J, Ciaccioni S. Olympic combat sports and mental health in children and adolescents with

disability: A protocol paper for systematic review. *PLoS ONE. 2025;20(2):e0301949.* DOI: https://doi.org/10.1371/journal.pone.0301949

[P8] Sung E*, **Lee Y***, Lee S, Park J. (*co-first authors) Analysis of sleep duration, energy intakes, physical activity, and metabolic syndrome based on the presence or absence of obesity and hypertension in working Korean adults. Frontiers in Public Health. 2025;13:1588706. *https://doi.org/10.3389/fpubh.2025.1588706*

[P7] Ciaccioni S, **Lee Y**, Guidotti F, Stankovic N, Pocscco E, Izzicupo P, Capranica L. Combat sports and wellbeing: Advancing health and inclusion in athletes and practitioners. *Frontiers in Psychology. 2025;16:1587672.* DOI: https://doi.org/10.3389/fpsyg.2025.1587672

[P6] Kim Y, Kim E, **Lee Y**, Park J. The role of late eating in circadian disruption and depression: A review of emotional health impacts. *Physical Activity and Nutrition. 2025;29(1):18-24.* DOI: https://doi.org/10.20463/pan.2025.0003

[P5] **Lee Y**, Kim N, Go S, Kim J, Park J. Sugary snack restriction enhances body composition improvement in overweight women engaging in non-face-to-face walking during COVID-19. *Frontiers in Public Health. 2024;12:1396598.* DOI: https://doi.org/10.3389/fpubh.2024.1396598

[P4] **Lee Y**, Hauck J, Brooks D. Sex-specific physical activity levels and energy intake in US young adults with depression: National Health and Nutrition Examination Survey 2017–2020 pre-pandemic data. *Physical Activity and Health. 2023;7(1):332-344.* DOI: https://doi.org/10.5334/paah.305

[P3] **Lee Y**, Park Y, Lee J, Sung E, Lee H, Park J. Household-specific physical activity levels and energy intakes according to the presence of metabolic syndrome in Korean young adults: Korean National Health and Nutrition Examination Survey 2016–2018. BMC Public Health. 2022;22:476. DOI: https://doi.org/10.1186/s12889-022-12852-3

[P2] Sung E, Lee S, **Lee Y**, Lee S, Park J. Analysis of energy intakes, physical activities and metabolic syndrome according to the income level in Korean elderly people: Korean National Health and Nutrition Examination Survey 2016–2018. *Physical Activity and Nutrition. 2022;26(2):28-35.* DOI: https://doi.org/10.20463/pan.2022.0011

[P1] Park Y, Lee J, **Lee Y**, Park J. Comparative analysis of energy intake and physical activity according to household type and presence of metabolic syndrome in middle-aged men based on data from the 7th Korea National Health and Nutrition Examination Survey (KNHANES) (2016–2018). *Physical Activity and Nutrition. 2021;25(4):1-9.* DOI: https://doi.org/10.20463/pan.2021.0020

Manuscripts Under Review – Under Revision

[UR4] **Lee Y**, Vazou S, Kim S, Kim N, Kim Y, Hauck J. A Systematic Review of Immersive VR Exergames for Individuals with Developmental Disabilities. *Disability and Rehabilitation: Assistive Technology.* Under review (2nd major revision).

[UR3] **Lee Y**, Chun H, Kim N, Stubbs B. Nonlinear Association Between Sedentary Time and Depression in the United States: Evidence from the 2021–2023 National Health and Nutrition Examination Survey. *Mental Health and Physical Activity.* Under review (2nd major revision).

[UR2] **Lee Y**, Brooks D, Kim S, Hauck J. Organized Sports and Psychosocial Benefits for Adolescents with Autism: Evidence from the 2022 National Survey of Children's Health. *European Journal of Adapted Physical Activity.* Under review (1st major revision).

[UR1] **Lee Y.** Socioeconomic Hardship and Cumulative Adverse Experiences in Relation to Anxiety and Depression in Autistic Children: Exploring the Role of Physical Activity. *Journal of Autism and Developmental Disorders.* Under review (2nd major revision).

Preprints

[PP3] **Lee Y**, Vazou S, Kim S, Ratan R, Hauck J. Interest in immersive virtual reality boxing exercise among college students with and without autism spectrum disorder. Research Square. 2026 DOI: https://doi.org/10.21203/rs.3.rs-9246183/v1

[PP2] **Lee Y**, Vazou S, Kim S, Hauck J. Effects of a 6-Week Virtual Boxing on Depressive Symptoms and Heart Rate Variability in Adults: A Pilot Randomized Controlled Trial. Research Square. 2026 DOI: https://doi.org/10.21203/rs.3.rs-9707388/v1

[PP1] **Lee Y**, Brooks D, Kim S, Hauck J. Organized Sports and Psychosocial Benefits for Adolescents with Autism: Findings from the 2022 National Survey. Research Square. 2025. DOI: https://doi.org/10.21203/rs.3.rs-7349284/v1

Open Science & Preregistrations

[O2] **Lee Y.** Adolescent Device-Measured Physical Activity and Early-Adult Psychological Distress in Autistic Young People: A Secondary-Data Preregistration (Millennium Cohort Study Robustness Analysis and Independent-Cohort Replication). *OSF Registries (Secondary Data Preregistration).* 2026. Embargoed until October 2028. https://osf.io/scugj

[O1] **Lee Y.** Wearable-Based Detection and Prediction of Depression in Autistic Individuals Using Physical Activity and Sleep Data: A Scoping Review. *OSF Preregistration.* 2026. DOI: https://doi.org/10.17605/OSF.IO/97DEK

PRESENTATIONS

Published Abstracts

[PA5] Lee J, **Lee Y**, Lin P, Luo T, Bonner S (June 2025, Poster). MoodMiles: A gamified exercise application for mental and physical well-being in young adults with all abilities. *Human-Computer Interaction International, Gothenburg, Sweden.*

[PA4] **Lee Y**, Ratan R, Vazou S, Kim S, Kim Y, Hauck J (June 2025, Poster). Exploring the effects of immersive virtual reality exercise on depressive symptoms, heart rate variability, and motivation for physical activity in young adults. *NASPSPA, Lake Tahoe, United States.*

[PA3] Brooks D, **Lee Y**, Bullard L, Hauck J, Vazou S (June 2025, Poster). Exploring contextual influences on school-based physical activity programs for youth with disabilities: A scoping review. *NASPSPA, Lake Tahoe, United States.*

[PA2] **Lee Y**, Capranica L, Pesce C, Guidotti F, Benzing V, Hauck J, Ciaccioni S (June 2024, Poster). Exploring the impact of Olympic combat sports on mental health of young individuals with disabilities: A meta-review protocol with preliminary insights. *NASPSPA, Louisiana, United States.*

[PA1] **Lee Y**, Brooks D, Hauck J (June 2023, Poster). Physical activity and energy intake in US young adults with depression: National Health and Nutrition Examination Survey 2017–2020 pre pandemic. *NASPSPA, Toronto, Canada.*

Conference Presentations

[CP13] **Lee Y,** Nye LM, Chen PW, Mitchell JA. Wearable-Derived Sleep and Physical Activity Levels and Rhythms and Depressive Symptoms in Adults with and without Autism Spectrum Disorder. Annual Meeting of the Associated Professional Sleep Societies, LCC (APSS), *2026, Baltimore, MD.*

[CP12] **Lee Y**, Nye LM, Chen PW, Bućan M, Mitchell JA (May 2026, Poster). Wearable-Derived Sleep and Physical Activity Levels and Rhythms and Depressive Symptoms in Adults with and without Autism Spectrum Disorder. Lurie Autism Institute Annual Symposium, Philadelphia, PA, United States.

[CP11] Brooks B, **Lee Y**, Hauck J, Fisher M, West P (Feb 2026, Poster). Health and the role of employment among young adults with intellectual and developmental disabilities: Scoping review. *MSU College of Nursing Research Day, Michigan, United States.*

[CP10] Brooks B, **Lee Y**, Hauck J, Fisher M, West P (Jan 2026, Poster). Health and the

role of employment among young adults with intellectual and developmental disabilities: Scoping review. *DADD, California, United States.*

[CP9] **Lee Y**, Kim S (July 2025, Poster). Synergizing physical activity and organized sports: Enhancing psychosocial well-being in adolescents with ASD in post-pandemic. *World Congress of Sociology of Sport, Seoul, Korea.*

[CP8] Kim S, **Lee Y**, Park S (July 2025, Poster). How team sports shape health in youth with autism: Examining the mediating effects of flourishing and socialization. *World Congress of Sociology of Sport, Seoul, Korea.*

[CP7] Brooks D, Hauck J, **Lee Y** (Sep 2024, Oral). Exploring physical activity experiences in children with autism spectrum disorder. *NAFAPA, New York, United States.*

[CP6] **Lee Y**, Brooks D, Hauck J (Sep 2024, Oral). Assessing the impact of an immersive virtual reality exercise on depression and anxiety in young adults with ASD. *NAFAPA, New York, United States.*

[CP5] Ciaccioni S, **Lee Y**, Hauck J, Capranica L, Guidotti F, Benzing V, Pesce C (July 2024, Oral). Olympic combat sports and mental health in children and adolescents with disability – preliminary results of a systematic review. *ECSS, Glasgow, Scotland.*

[CP4] Anttila E, Brooks D, **Lee Y**, Hauck J (April 2023, Poster). Examining the differences in PA levels between unified partners and Special Olympic athletes at the 2022 Special Olympics Unified Cup Soccer. *UURAF, Michigan, United States.*

[CP3] Go S, **Lee Y**, Kim N, Park J (Aug 2022, Oral). The effects of non-face-to-face intervention of brisk walking and sugar restriction on body weight, physical fitness and insulin sensitivity in overweight and obese men. *KAHPERD, Gyeonggi-do, Korea.*

[CP2] **Lee Y**, Kim N, Go S, Park J (Mar 2022, Oral). Effect of non-contact brisk walking and sugar intake restriction on body fat and osteocalcin (OC) in young women during the COVID-19 outbreak period. *KSSO, Seoul, Korea.*

[CP1] **Lee Y**, Kim N, Go S, Park J (Oct 2021, Oral). Effect of non-contact brisk walking and sugar intake restriction on body fat and osteocalcin (OC) in young single-person households during the COVID-19 outbreak period. *KAHPERD, Gumi, Korea.*

Invited Lectures & Keynote Presentations

[L1] **Lee Y** (Mar 2026, Oral). Bridging the Gap: Immersive Virtual Reality as a New Pathway for Physical Activity and Mental Health in Autism. *World Autism Awareness Day Symposium, East China Normal University, Shanghai, China.*

RESEARCH GRANTS

Awarded

- KIN Endowed Fellowship, Michigan State University (2025), $1,500
- Future of Work Graduate Student Funding, Michigan State University (2025), PI, $3,000
- Summer Research Fellowship, Michigan State University (2025), $7,000
- Hard Cost Practicum/Dissertation Funding, Michigan State University (2024), PI, $1,763
- Summer Research Fellowship, Michigan State University (2024), $7,000
- Summer Research Development Fellowship, Michigan State University (2023), $7,000
- Summer Research Development Fellowship, Michigan State University (2022), $6,000

TEACHING EXPERIENCE

Graduate Teaching Assistant, Michigan State University (2022 – 2025)

- KIN 492 Internship in Kinesiology
- KIN 300C Soccer Coaching

- KIN 173 Foundations of Kinesiology
- KIN 121 The Healthy Lifestyle
- KIN 116B Golf
- KIN 108R Indoor Soccer
- KIN 107E Tennis
- KIN 103T Distance Running
- KIN 103G Power Walking
- KIN 103D General Conditioning
- KIN 103A Aerobic Exercise

TEACHING GRANTS

- Inclusive Teaching Grant, Michigan State University (2025), $1,400

SCHOLARSHIP AWARDS

- David Cho Scholarship Award, Michigan State University (2025), $1,000
- Dr. Kenneth Hardin Graduate Student Endowed Scholarship, Michigan State University (2025), $1,500
- Henry J. and Betty Montoye Endowed Scholarship Award, Michigan State University (2024), $2,325

TRAVEL AWARDS

- Travel Fund, Department of Kinesiology, Michigan State University (2024), $600
- COGS Conference Award, Michigan State University (2024), $300
- Travel Fund, Department of Kinesiology, Michigan State University (2024), $400
- Travel Fund, Graduate School, Michigan State University (2023), $600
- Travel Fund, Department of Kinesiology, Michigan State University (2023), $400

MENTORSHIP ACTIVITIES

Mentored Undergraduate Students in Journal Club

Ronnie Aniel, Anela Panlasigue, Sathvik Suryadevara, Emma Anttila, Monique Karoub, Elaine Nevins, Brooke Johnson, Chloe Barnes, Rylee Bernhauser, Landon Moreno, Bronwen Smith, Christopher Stubbe, Laura Whitney, Maryann Hermiz, Alaina Barringer, Sarah Segarra, Lydia Miller, Morgan Adelini

PROFESSIONAL AFFILIATIONS

- Associated Professional Sleep Societies (2026 – Present)
- North American Federation of Adapted Physical Activity (2024 – Present)
- North American Society for the Psychology of Sport and Physical Activity (2023 – Present)
- Korean Society Exercise Nutrition (2021 – Present)
- Korean Alliance for Health, Physical Education, Recreation, and Dance (2021 – Present)

SERVICE ACTIVITIES

Scholarly and Professional Organizations

- Student Ambassador, NASPSPA, Toronto, Canada (2023)
- Editorial Director, Korean Society of Exercise Nutrition, Seoul, Korea (2022)

Reviewing for Journals

- Games for Health Journal (2026 – Present)
- American Journal of Preventive Medicine (2026 – Present)
- BMC Sports Science (2026 – Present)
- Sports Sciences for Health (2026 – Present)
- Frontiers in Psychology (2026 – Present)
- Journal of Applied Research in Intellectual Disabilities (2025 – Present)
- Mental Health and Physical Activity (2025 – Present)
- BMJ Open (2025 – Present)
- Applied Psychology: Health and Well-Being (2025 – Present)
- Frontiers in Psychiatry (2025 – Present)

- Journal of Physical Activity and Health (2024 – Present)
- Journal for Advancing Sport Psychology In Research (2024 – Present)

MILITARY SERVICE

- Ulleungdo & Dokdo Security Unit, Gyeongbuk, Korea (2016 – 2017), Mandatory Service as a Police Officer

MEDIA & PRESS COVERAGE

- WKAR Public Media (October 21, 2025). *From the ring to the MSU classroom: How VR technology is helping with autism spectrum disorder.* Link

SKILLS

- Expertise in analyzing large-scale population health datasets (e.g., NHANES, NSCH, HBSC, KYRBS, All of Us Research Program) for physical activity epidemiology
- Experienced in processing and analyzing wearable-derived behavioral data (e.g., Fitbit, GENEActiv, ActiGraph) for physical activity, sleep, and circadian rhythm metrics
- Proficient in statistical analysis using R and SPSS
- Experienced in designing and implementing immersive virtual reality (VR)-based exercise interventions
- Experienced in biological and physiological assessments, including blood biomarker analysis (e.g., glucose, insulin, osteocalcin) and heart rate variability (e.g., RMSSD, SDNN, LF/HF ratio)
- Experienced in conducting systematic and scoping reviews aligned with PRISMA guidelines

Last Updated: Oct 2026